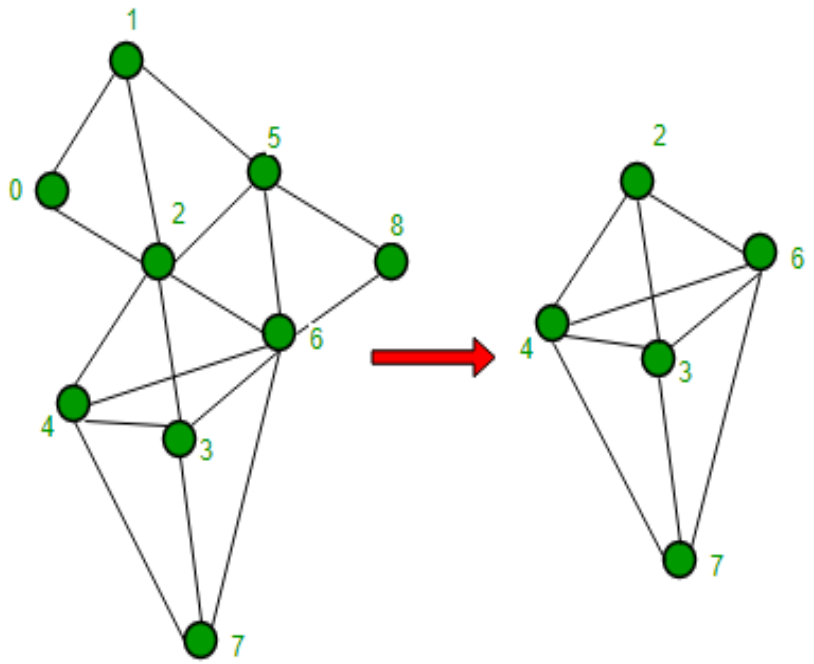
Exercise#1:

Given a graph G and an integer K, K-cores of the graph are connected components that are left after all vertices of degree less than k have been removed.



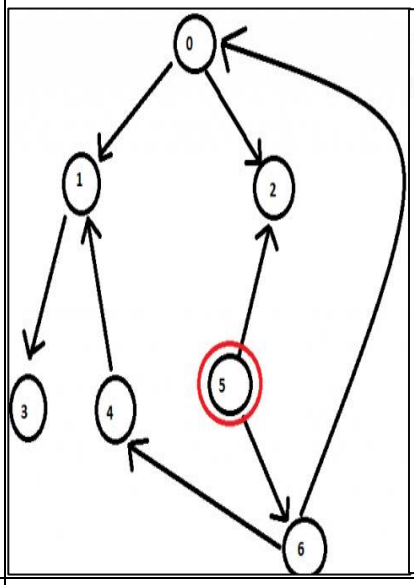
3-Cores

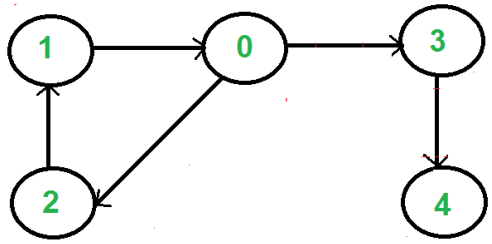
Input Graph

Example 1 :	
Input	Adjacency list representation of graph shown in the left diagram above. K = 3
Output	{2: [3,4,6], 3: [2,4,6,7], 4: [2,3,6,7], 6: [2,3,4,7], 7: [3,4,6] }

Exercise # 2

Find a Mother vertex in a graph. Mother vertex in a graph G = (V, E) is a vertex v such that all other vertices in G can be reached by a path from v.

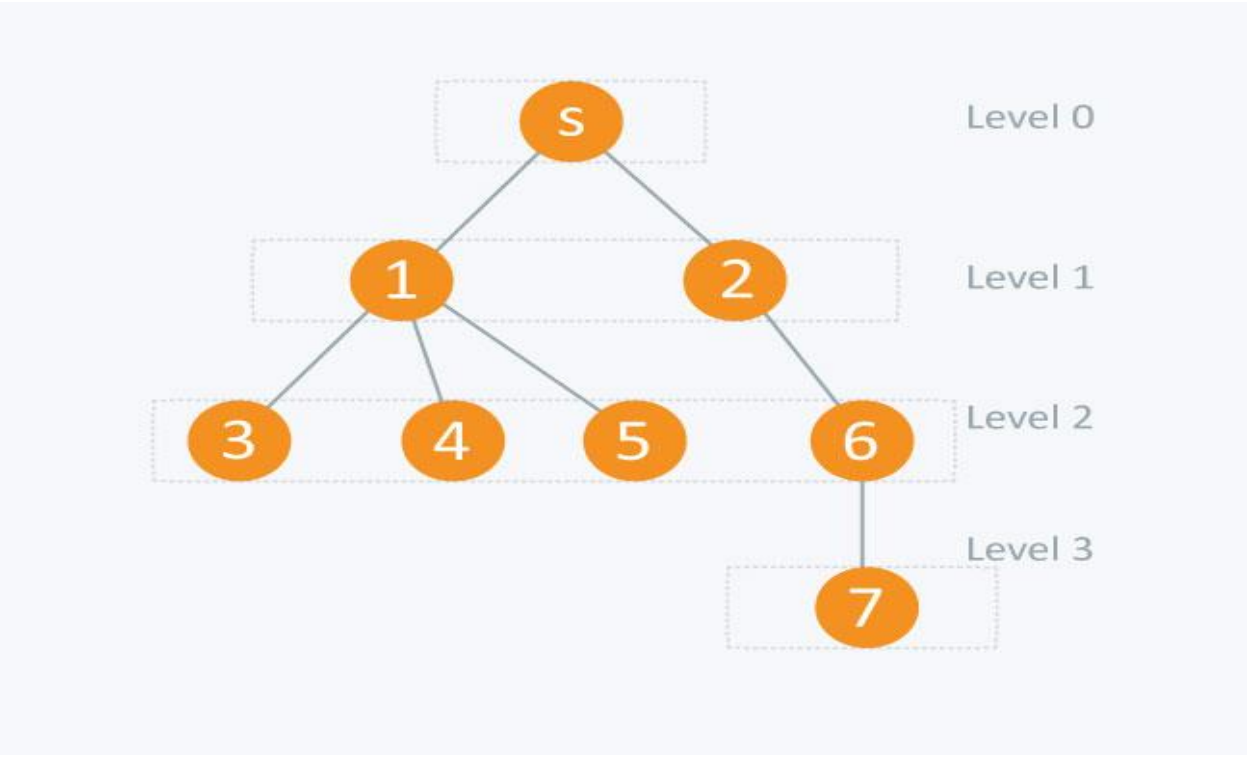
Example 1:	
Input	Adjacency list of the graph shown below  In this graph the mother vertex is- '5'(circled red) as we can reach any node from - '5' through a directed path *To reach 0-* *5->6->0* *To reach 1-* *5->6->0->1* *To reach 2-* *5->2* *To reach 3-* *5->6->0->1->3* *To reach 4-* *5->6->4* *To reach 6-* *5->6*
Output	[5]

Example 2:	
Input	Adjacency list representation of a graph shown below  Mother Vertices : 0, 1 and 2
Output	[0, 1, 2]

Exercise#3: Level of each node in the graph

How to determine the level of each node in the given graph?

As you know in BFS, you traverse level wise. You can also use BFS to determine the level of each node.



Exercise# 3b:

Write a function get_node_level(G, node) that takes a graph and node as input and returns its level.

Example 1 :	
Input	Adjacency list representation of graph shown above. node = 2
Output	1

Example 2 :	
Input	Adjacency list representation of graph shown above. node = 6
Output	2

Exercise#4: Course Schedule

Bob loves foreign languages and wants to plan his course schedule for the following years.

He is interested in the following nine language courses: LA15, LA16, LA22, LA31, LA32, LA126, LA127, LA141, and LA169.

The course prerequisites are:

- LA15: (none)
- LA16: LA15
- LA22: (none)
- LA31: LA15
- LA32: LA16, LA31
- LA126: LA22, LA32
- LA127: LA16
- LA141: LA22, LA16
- LA169: LA32

In what order can Bob take these courses, respecting the prerequisites?

Example 1 :	
Input	List of Courses & Course Pre-Requisites
Output	There can be different outputs/course plan such as: • LA15, LA22, LA16, L31, L32, LA169, LA126, LA127, LA141 • LA15, LA22, LA16, LA31, LA32, LA127, LA141, LA126, LA169 • LA15, LA16, LA22, LA31, LA32, LA126, LA127, LA141, LA169